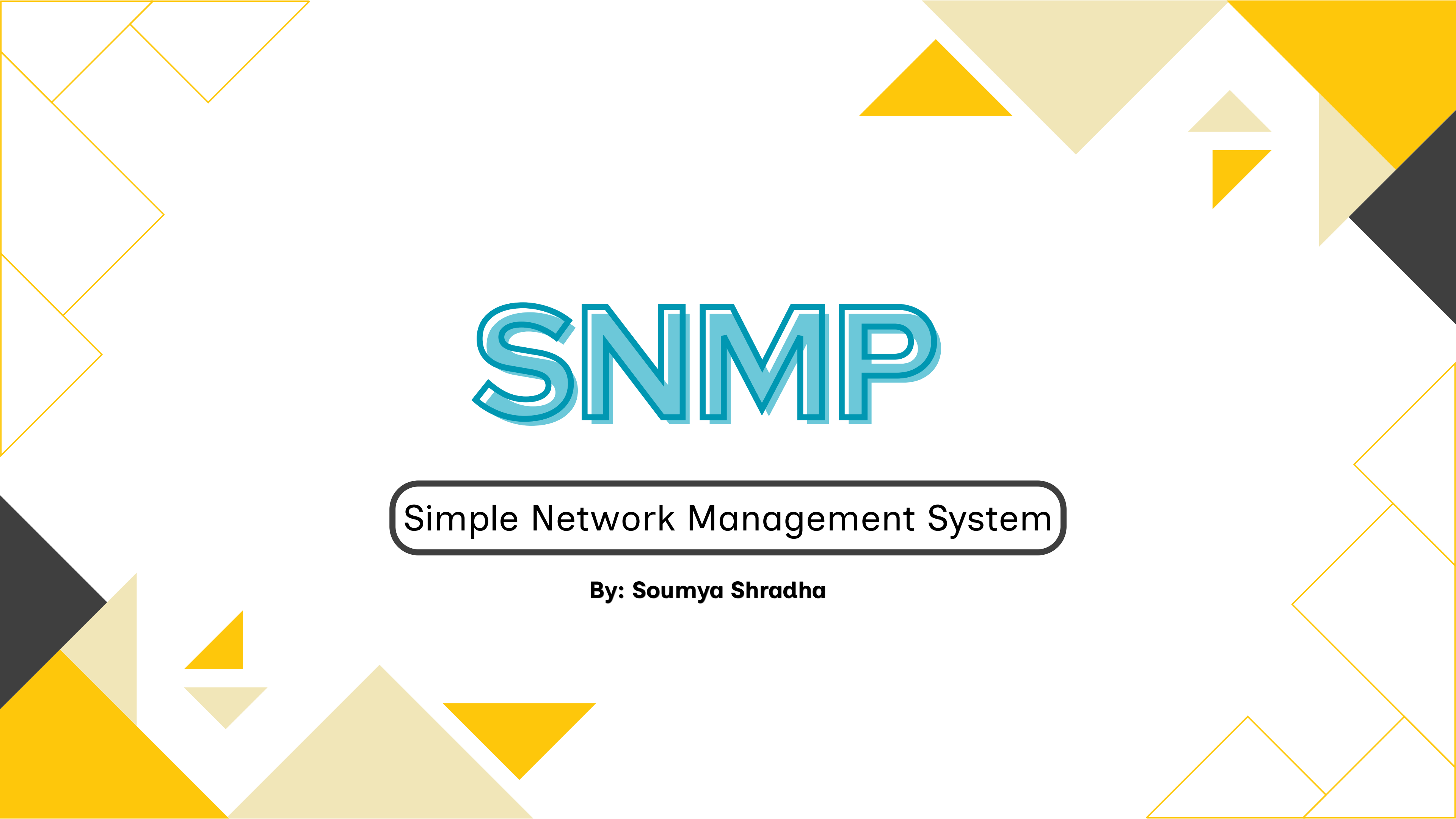




SNMP

Simple Network Management System

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Simple Network Management Protocol

Simple Network Management Protocol (SNMP) is a widely used protocol for network management that provides a standardized framework for monitoring and managing network-connected devices in IP networks such as: **routers, switches, servers, printers, firewalls, and load balancer.**

It works at the application layer of the Internet Protocol suite

It helps network administrators to:

- Monitor how well the network is performing.
- Detect and fix network problems.
- Plan future expansions of the network.
- Sometimes control or configure devices remotely.

Need Of SNMP

It was created to solve the problem of managing growing networks as the Internet started expanding,

- SNMP was first introduced in- 1988.
- then in 1990-**SNMPv1** officially defined and accepted as a standard (RFC 1157).
- SNMP communicates using **UDP** ports 161 and 162.
- SNMP uses UDP to send and receive messages-
 - 1)**Port 161**-> Used for requests and responses
(Manager ↔ Agent communication like GET, SET)
 - 2)**Port 162**->Used for TRAP messages (alerts).
(Agent → Manager, when something goes wrong)

Why SNMP uses UDP because?

- It is faster(no connection setup like TCP)
- It has low overhead.
- Suitable for simple,quick communication.

But:

- it is less reliable(no gurantee of delivery)

Key Components Of SNMP

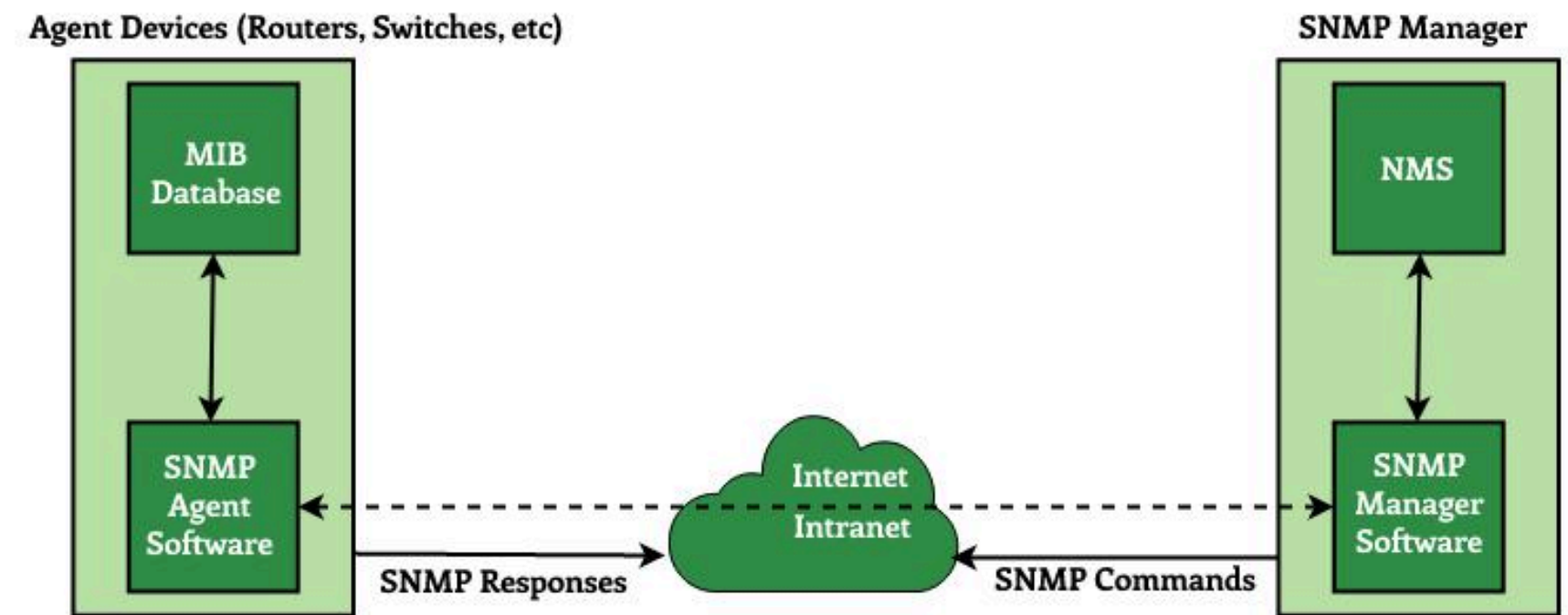
- 1)SNMP Manager
- 2)SNMP Agent
- 3)Management Information Base (MIB)

Architecture Of SNMP

SNMP Architecture

1) SNMP Manager (Brain of the system)

- It is a centralized system used to monitor the network. It is also known as a Network Management Station (NMS).
- A router that runs the SNMP server program is called an agent, while a host that runs the SNMP client program is called a manager.
- It monitors all devices in the network.
- It sends requests and receives responses.



2) SNMP Agent (Helper inside devices)

- This is a software installed on network devices like routers, switches, etc.
- The agent stores network data in the MIB, and the manager accesses this data to monitor the network. For example, it can check packet counts to see if a router is congested

Architecture Of SNMP

3) MIB (Management Information Base)

This is like a database or storage.

- It stores all information about the device
- Data is stored in the form of variables (objects)
- Organized in a tree-like structure (hierarchy)

Example data:

- 1)CPU usage
- 2)Network traffic
- 3)IP details

Working...

Step 1: Manager sends request

Step 2: Agent collects data from MIB

Step 3: Agent sends response back

Step 4: Manager displays it

SNMP Messages

1) GetRequest: (Used by Manager → Agent)

- To get data from a device
- Agent sends back a Response

2) GetNextRequest: (Used by Manager → Agent)

- To get the next value in a list/table
- Useful when manager doesn't know exact index

3) SetRequest: (Used by Manager → Agent)

- It is used by the SNMP manager to set the value of an object instance on the SNMP agent.

4) Response: (Used by Agent → Manager)

- sent in response to the Set message, it will contain the newly set value as confirmation that the value has been set.

5) Trap: (Used by Agent → Manager(without request))

- These are the message sent by the agent without being requested by the manager. It is sent when a fault has occurred.

6) InformRequest: (Added to SNMPv2c)

- Used to determine if the manager has received the trap message or not.
- It is the same as a trap but adds an acknowledgement that the trap doesn't provide.

SNMP Security Levels

1) noAuthNoPriv—This (no authentication, no privacy) security level uses a community string for authentication and no encryption for privacy.

2) AuthNoPriv—This (authentication, no privacy) security level uses HMAC with MD5 for authentication and no encryption for privacy.

3) authPriv—This security level (authentication, privacy) uses HMAC with MD5 or SHA for authentication and encryption uses the DES-56 algorithm.

Versions Of SNMP

SNMPv1 (Basic Version) (1990)

- First version of SNMP
- Simple and easy to implement
- Uses community string (password)
- Uses UDP protocol
- No strong security

SNMPv2c (Improved Version) (1996)

- Improved version of SNMPv1
- Still uses community string (weak security)
- Faster and more efficient
- Uses UDP (mainly)

Improvements:

- Better MIB handling
- More message types

SNMPv3 (Secure Version) (1998)

- Most advanced version
- Provides authentication (MD5/SHA)
- Provides encryption (DES)
- Ensures secure communication by using authentication
- allows administrators to safely configure network devices remotely.

Characteristics

- SNMP is used to monitor network.
- It detects any network faults.
- It can also be used to configure remote devices.
- It allows a standardized way of collecting information about all kinds of devices from various manufacturers among the networking industry.

Advantages

- It is easy to implement.
- Agents are widely implemented.
- Agent level overhead is minimal.
- It is robust and extensible.
- Polling approach is good for LAN based managed object.
- It offers the best direct manager agent interface.

Limitations

- It does not scale well.
- There is no object oriented data view.
- It has no standard control definition.
- It has many implementation specific (private MIB) extensions.
- It has high communication overhead due to polling



Thank You